

Problem set 12 - Due: Wednesday, April 29

1. A body with mass $m = \frac{1}{2}$ kilogram (kg) is attached to the end of a spring stretched 2 meters (m) by a force of 100 newtons (N). It is set in motion with initial position $x_0 = 1(m)$ and initial velocity $v_0 = 5(m/s)$. (Note that these initial conditions indicate that the body is displaced to the right and is moving to the left at time $t = 0$.) Derive the equation of motion and find the position function of the body.

We have

$$100 = k(2) \implies k = 50,$$

with no damper and no external force. Thus, the ODE is

$$\frac{1}{2}x'' + 50x = 0, \quad x(0) = 1, \quad x'(0) = -5\frac{m}{s}.$$

We have the C.E

$$\frac{1}{2}r^2 + 50 = 0 = r^2 + 100.$$

So,

$$r = \pm 10i.$$

Thus,

$$\begin{aligned} x(t) &= c_1 \cos(10t) + c_2 \sin(10t), \\ x'(t) &= -10c_1 \sin(10t) + 10c_2 \cos(10t). \end{aligned}$$

With the initial conditions

$$\begin{aligned} 1 &= c_1 \cos(0) + c_2 \sin(0) = c_1, \\ -5 &= -10(1) \sin(0) + 10c_2 \cos(0) = 10c_2. \end{aligned}$$

Thus, $c_1 = 1$, $c_2 = -\frac{1}{2}$, and the equation is therefore

$$x(t) = \cos(10t) - \frac{1}{2} \sin(10t).$$

2. An object stretches a spring 6 inches in equilibrium. Find its displacement for $t > 0$ if it is initially displaced 18 inches above equilibrium and given a downward velocity of 3ft/s

We have

$$mg = kx \implies k = \frac{mg}{x} = \frac{mg}{6 \text{ in}} = \frac{mg}{0.5 \text{ ft}} = 2mg.$$

With

$$mx'' + kx = 0,$$

we get

$$x'' + \frac{k}{m}x = 0 = x'' + \frac{2mg}{m}x.$$

Thus,

$$x'' + 2gx = 0.$$

Using $g = 32 \text{ ft/s}^2$,

$$x'' + 64x = 0,$$

With initial conditions $x_0 = 18 \text{ in} = \frac{3}{2} \text{ ft}$, $x'(0) = -3 \text{ ft/s}$. The characteristic equation is

$$r^2 + 64 = 0.$$

Thus, $r = \pm 8i$, and

$$\begin{aligned} x(t) &= c_1 \cos(8t) + c_2 \sin(8t), \\ x'(t) &= -8c_1 \sin(8t) + 8c_2 \cos(8t). \end{aligned}$$

Thus,

$$\frac{3}{2} = c_1 \cos(0) + c_2 \sin(0) = c_1,$$

and

$$\begin{aligned} -3 &= -8 \left(\frac{3}{2} \right) \sin(8t) + 8c_2 \cos(8t) \\ &= -12 \sin(0) + 8c_2 \cos(0) = 8c_2. \end{aligned}$$

Thus, $c_1 = \frac{3}{2}$ and $c_2 = -\frac{3}{8}$. With these constants, we have

$$x(t) = \frac{3}{2} \cos(8t) - \frac{3}{8} \sin(8t).$$